Cloud Computing

Mock Exam 4 | Course ID: noc26_cs55

Q1. What does XaaS stand for in cloud computing?

A) eXtra as a Service
B) Anything as a Service
C) eXtreme as a Service
D) All as a Service

Source: Lecture_W1L19_Lecture_1__Cloud_Computing___Overview | Difficulty: Easy

Q2. Which cloud service model delivers ready-to-use software applications over the web, typically via a web browser?

A) IaaS (Infrastructure as a Service)
B) PaaS (Platform as a Service)
C) SaaS (Software as a Service)
D) XaaS (Anything as a Service)

Source: Lecture_W1L21_Lecture_3__Cloud_Computing___Introduction | Difficulty: Easy

Q3. What is the primary method for achieving elasticity in the cloud by adding more machines (nodes) to a resource pool to increase capacity?

A) Vertical Scaling (Scale-Up)
B) Horizontal Scaling (Scale-Out)
C) Diagonal Scaling
D) Static Scaling

Source: Lecture_W1L22_Lecture_4__Cloud_Computing_Architecture | Difficulty: Easy

Q4. What is the core technology that differentiates cloud computing from the traditional client-server model, enabling multi-tenancy and dynamic resource allocation?

A) Dedicated Servers
B) Virtualization
C) Fixed IP Addressing
D) Local Storage

Source: Lecture_W1L23_Lecture_5__Cloud_Computing_Architecture__contd___ | Difficulty: Easy

Q5. Which cloud deployment model is provisioned for open use by the general public, owned and operated by a third-party provider (e.g., AWS, Azure, Google Cloud)?

A) Private Cloud
B) Community Cloud
C) Public Cloud
D) Hybrid Cloud

Source: Lecture_W2L28_Lecture_6__Cloud_Computing__Architecture___Deploym | Difficulty: Easy

Q6. What is the software layer that creates and runs virtual machines (VMs) and manages the allocation of physical resources to each VM?

A) Guest OS
B) Host OS
C) Hypervisor (VMM)
D) Application Layer

Source: Lecture_W2L29_Lecture_7__Cloud_Computing__Virtualization | Difficulty: Easy

Q7. What is the minimum requirement for an XML document to be processed by any XML parser, ensuring all tags are closed, properly nested, and there is one root element?

A) Valid XML
B) Schema-compliant XML
C) Well-Formed XML
D) DTD-compliant XML

Source: Lecture_W2L30_Lecture_8__Cloud_Computing__XML_Basics | Difficulty: Easy

Q8. What is a formal, legally binding contract between a service provider and a service consumer that defines the level of service, responsibilities, and remedies for non-compliance?

A) Key Performance Indicator (KPI)
B) Service Level Objective (SLO)
C) Service Level Agreement (SLA)
D) Quality of Service (QoS)

Source: Lecture_W3L37_Lecture_11__Service_Level_Agreement | Difficulty: Easy

Q9. What economic principle states that cost per unit decreases as the scale of operation increases, often seen in large cloud providers?

A) Utility Pricing
B) Statistical Multiplexing
C) Economies of Scale
D) Coefficient of Variation

Source: Lecture_W3L38_Lecture_12__Cloud_Economics | Difficulty: Easy

Q10. In Docker, what is a lightweight, standalone, executable package that includes everything needed to run a piece of software: the code, runtime, system tools, libraries, and settings?

A) Virtual Machine
B) Image
C) Hypervisor
D) Kernel

Source: Lecture_W8L82_Lecture_36___Introduction_to_DOCKER_Container | Difficulty: Easy

Q11. Which computing paradigm introduced the concept of tapping into a vast pool of computing power like a utility, often harnessing underutilized compute cycles, and is analogous to an 'electrical grid'?

A) Distributed Systems
B) Cluster Computing
C) Grid Computing
D) Utility Computing

Source: Lecture_W1L20_Lecture_2__Cloud_Computing___Overview__contd___ | Difficulty: Medium

Q12. Which NIST essential characteristic of cloud computing ensures that capabilities can be elastically provisioned and released—in some cases automatically—to scale rapidly outward and inward commensurate with demand?

A) On-Demand Self-Service
B) Broad Network Access
C) Resource Pooling
D) Rapid Elasticity

Source: Lecture_W1L21_Lecture_3__Cloud_Computing___Introduction | Difficulty: Medium

Q13. Which XML parser builds a complete in-memory tree representation of an entire XML document, allowing for random access, complex queries, and dynamic modification of the

document?

A) SAX Parser
B) DOM Parser
C) XSLT Processor
D) XPath Engine

Source: Lecture_W2L31_Lecture_9__Cloud_Computing__XML_Basics_II | Difficulty: Medium

Q14. In Service Oriented Architecture (SOA), which component acts as the 'Yellow Pages' or directory for web services, allowing service providers to publish their services and for consumers to find them?

A) XML
B) SOAP
C) WSDL
D) UDDI

Source: Lecture_W2L32_Lecture_10__Cloud_Computing__Web_Services__Service | Difficulty: Medium

Q15. Which distributed file system, designed to store massive files (terabytes) across large clusters of commodity hardware, provides the foundational storage layer for many cloud data systems and is often implemented as HDFS?

A) NTFS
B) FAT32
C) Google File System (GFS)
D) ext4

Source: Lecture_W3L39_Lecture_13__Managing_Data | Difficulty: Medium

Q16. In MapReduce, what is the primary purpose of the Reducer task?

A) To split the input data into smaller chunks.
B) To apply a user-defined function to each data chunk and emit key-value pairs.
C) To receive intermediate key-value pairs, group them by key, and aggregate the values.
D) To manage fault tolerance and task scheduling.

Source: Lecture_W3L40_Lecture_14__Introduction_to_Map_Reduce | Difficulty: Medium

Q17. In OpenStack, which service is the primary compute engine, responsible for managing the entire lifecycle of virtual machines (VMs), including spawning, scheduling, and decommissioning?

A) Horizon
B) Neutron
C) Nova
D) Keystone

Source: Lecture_W3L41_Lecture_15___Open_Stack | Difficulty: Medium

Q18. In Microsoft Azure's architectural blueprint, which computational role is designed for running background processing, distributed tasks, or other complex, non-interactive computations?

A) Web Role
B) Front-End Role
C) Worker Role
D) Database Role

Source: Lecture_W4L47_Lecture_17___Cloud_Computing__Case_Study_with_a_Co | Difficulty: Medium

Q19. When calculating the average of a set of integers using MapReduce, if mappers emit local averages and counts, why can't the reducer simply average the incoming local averages?

A) Because the reducer only processes sums, not averages.
B) Because the local averages might be based on different numbers of elements.
C) Because the reducer needs to sort the data first.
D) Because the local averages are not weighted.

Source: Lecture_W5L57_Lecture_23__MapReduce_Tutorial | Difficulty: Medium

Q20. According to the lecture, what are the three goals of security management in any complex system?

A) Identification, Authentication, Authorization
B) Prevention, Detection, Recovery
C) Confidentiality, Integrity, Availability
D) Policy, Mechanism, Trust

Source: Lecture_W6L64_Lecture_26__Cloud_Computing__Security_I | Difficulty: Medium

Q21. What is the core mechanism of Mobile Cloud Computing (MCC) that involves migrating data-intensive or computationally heavy parts of an application from a mobile device to a resourceful server in the cloud for execution?

A) Data Caching
B) Computation Offloading
C) Local Processing
D) Device Synchronization

Source: Lecture_W7L73_Lecture_31__Mobile_Cloud_Computing__I | Difficulty: Medium

Q22. In Sensor Cloud, what is a software-based emulation of a physical sensor that obtains its data from one or more underlying physical sensors, providing a customized and transparent view to users and applications?

A) Physical Sensor (PS)
B) Sensor Cloud Proxy
C) Virtual Sensor (VS)
D) Derived Sensor

Source: Lecture_W8L84_Lecture_38___Sensor_Cloud_Computing | Difficulty: Medium

Q23. In the proposed Coordinated Voltage Control (CVC) scheme for AC-DC hybrid smart grids, what is the strategic reason for introducing an intentional time delay on fast-acting devices (like DSTATCOM) during steady and dynamic states?

A) To allow the system to cool down and prevent overheating.
B) To give slow-acting devices (like OLTC) enough time to contribute to voltage correction and preserve reactive power reserves of fast devices.
C) To increase the overall response time of the system for better stability.
D) To simplify the control logic by reducing simultaneous actions.

Source: Lecture_W10L102_Lecture_48___AC-DC_Smart_Grid_Case_Study__Coordinated_Voltage_Control | Difficulty: Hard

Q24. According to the experimental findings presented in the lecture, what was the Total Harmonic Distortion (TSD) of the source current reduced to after compensation by the Unified Power Quality Conditioner (UPQC), bringing it well below the IEEE 519 standard of 5%?

A) 26.19%
B) 10.5%
C) 5.0%
D) 3.1%

Source: Lecture_W10L104_Lecture_50___Docker_Container____Demo__Part_II_ | Difficulty: Hard

Q25. In an isolated DC microgrid operating in surplus power mode with a full battery (SOC is not full), what action is taken by the Power Flow Control Strategy (PFCS) to prevent system instability?

A) The microgrid exports excess energy to the main grid.
B) The battery discharges to cover the surplus.
C) The PV system is commanded to operate in 'off-MPP' mode, reducing its output.
D) Non-essential loads are shed to consume the excess power.

Source: Lecture_W11L109_Lecture_51___Dew_Computing | Difficulty: Hard

Q26. What is the primary reason asymmetrical faults are significantly more disruptive to Doubly-Fed Induction Generator (DFIG) operation compared to symmetrical faults of the same voltage depth?

A) Asymmetrical faults cause higher voltage dips.
B) Asymmetrical faults introduce negative sequence components, leading to severe instability in rotating machines.
C) Symmetrical faults are easier for DFIG control systems to manage.
D) Asymmetrical faults only affect one phase, making them harder to detect.

Source: Lecture_W12L119_Lecture_57___CPS_and_Cloud_Computing | Difficulty: Hard

Q27. Which virtualization approach combines the performance of paravirtualization with the compatibility of full virtualization by leveraging special processor extensions (e.g., Intel VT-x, AMD-V) to run an unmodified guest OS efficiently?

A) Full Virtualization
B) Paravirtualization
C) Hardware-Assisted Virtualization
D) Containerization

Source: Lecture_W2L29_Lecture_7__Cloud_Computing__Virtualization | Difficulty: Hard

Q28. According to the lecture, what is the mathematical condition under which cloud computing is more economical than owning on-premise infrastructure, where P is peak demand, A is average demand, and U is the utility premium?

A) P < A * U
B) P / A > U
C) P + A < U
D) P * A = U

Source: Lecture_W3L38_Lecture_12__Cloud_Economics | Difficulty: Hard

Q29. According to the 'Hey, You, Get Off of My Cloud' paper, what is the primary goal of the 'placement' phase of a co-residence attack in a multi-tenant cloud environment?

A) To identify the target instance's public IP address.
B) To intentionally place a malicious VM on the same physical server as the victim's VM.
C) To map the internal network topology of the cloud provider.
D) To exploit side-channel information leakage from a remote server.

Source: Lecture_W6L66_Lecture_28__Cloud_Computing__Security_III | Difficulty: Hard

Q30. According to the mathematical view presented in the lecture, which condition is NOT explicitly beneficial for offloading a task in Mobile Cloud Computing?

A) The task is compute-intensive (C is large).
B) The cloud server is significantly faster (F is high).
C) The network bandwidth (B) is low.
D) The amount of data to transfer (D) is small.

Source: Lecture_W7L74_Lecture_32__Mobile_Cloud_Computing__II | Difficulty: Hard

Answer Key

1: B	2: C	3: B	4: B	5: C
6: C	7: C	8: C	9: C	10: B
11: C	12: D	13: B	14: D	15: C
16: C	17: C	18: C	19: B	20: B
21: B	22: C	23: B	24: D	25: C
26: B	27: C	28: B	29: B	30: C